

2011
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 9, 11, 13, 15, 17, 19 and 21

Total marks – 100

Section I Pages 2–22

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–32
- Allow about 1 hour and 40 minutes for this part

Section II Pages 23–33

25 marks

- Attempt ONE question from Questions 33–37
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

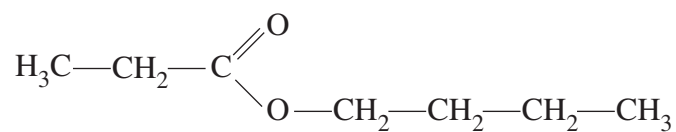
- 1 Which of the following industrial processes is used to produce ethanol from ethylene?
 - (A) Hydration
 - (B) Dehydration
 - (C) Addition polymerisation
 - (D) Condensation polymerisation

- 2 Which of the following shows two products that result from the fermentation of glucose?
 - (A) Cellulose and water
 - (B) Ethanol and oxygen
 - (C) Carbon dioxide and water
 - (D) Ethanol and carbon dioxide

- 3 Which of the following lists contains ONLY basic substances?
 - (A) Oven cleaner, urine, vinegar
 - (B) Lemonade, drain cleaner, blood
 - (C) Baking soda, ammonia, sea water
 - (D) Antacid, dishwashing detergent, lemon juice

- 4 Which of the following gases can cause major depletion of the ozone layer?
 - (A) O_2
 - (B) NO_2
 - (C) CO_2
 - (D) CCl_3F

- 5 Iron (III) chloride and aluminium sulphate are two chemicals that can be used in the purification of town water supplies. What is the role of these chemicals?
- (A) To disinfect water by removing bacteria
 (B) To remove particulate material by flocculation
 (C) To control the concentration of total dissolved solids
 (D) To control the pH of the water within the required range
- 6 Which property would be most useful in distinguishing between butan-1-ol and propan-1-ol?
- (A) Boiling point
 (B) Colour
 (C) Conductivity
 (D) Density
- 7 Which of the following lists contains ONLY unstable isotopes?
- (A) $^{207}_{82}\text{Pb}$, $^{99}_{43}\text{Tc}$, $^{12}_7\text{N}$
 (B) $^{214}_{82}\text{Pb}$, $^{46}_{20}\text{Ca}$, $^{99}_{43}\text{Tc}$
 (C) $^{238}_{92}\text{U}$, $^{40}_{20}\text{Ca}$, $^{12}_7\text{N}$
 (D) $^{238}_{92}\text{U}$, $^{40}_{20}\text{Ca}$, $^{99}_{43}\text{Tc}$
- 8 What is the systematic name of the molecule shown?

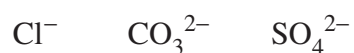


- (A) Butyl butanoate
 (B) Propyl butanoate
 (C) Butyl propanoate
 (D) Propyl propanoate

9 What property of O_3 makes it more soluble in water than O_2 in water?

- (A) O_3 is a polar molecule.
- (B) O_3 has a resonance structure.
- (C) O_3 is a highly reactive molecule.
- (D) O_3 has a coordinate covalent bond.

10 An aqueous sample containing the following anions is analysed.



In which order should the reagents be added to determine the amount of chloride in the sample?

	<i>Reagent 1</i>	<i>Reagent 2</i>	<i>Reagent 3</i>
(A)	$AgNO_3$	H_2SO_4	$BaSO_4$
(B)	HCl	$Pb(NO_3)_2$	$AgNO_3$
(C)	HNO_3	$Ba(NO_3)_2$	$AgNO_3$
(D)	$Ba(NO_3)_2$	$AgNO_3$	CH_3COOH

11 Which compound can form when bromine water reacts with propene?

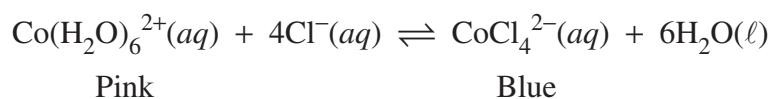
- (A) 1-bromopropane
- (B) 2-bromopropane
- (C) 1,1-dibromopropane
- (D) 1,2-dibromopropane

- 12 A table of redox couples and their standard reduction potentials is shown.

Redox couple	$E^\ominus$
Ag^+/Ag	0.80 V
Cd^{2+}/Cd	-0.40 V
Pd^{2+}/Pd	0.92 V
Ni^{2+}/Ni	-0.24 V

Which of the following ranks the metals in decreasing order of their electrochemical activity?

- (A) $\text{Ni} > \text{Cd} > \text{Ag} > \text{Pd}$
(B) $\text{Pd} > \text{Ag} > \text{Cd} > \text{Ni}$
(C) $\text{Pd} > \text{Ag} > \text{Ni} > \text{Cd}$
(D) $\text{Cd} > \text{Ni} > \text{Ag} > \text{Pd}$
- 13 When chloride ions are added to a solution containing $\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq})$, the following equilibrium is established.



Which of the following statements about the colour of the solution is true?

- (A) Diluting the solution with water will make it turn blue.
(B) If the reaction is exothermic, heating the solution will make it turn blue.
(C) If the reaction is endothermic, cooling the solution will make it turn pink.
(D) Adding a large amount of solid potassium chloride to the solution will make it turn pink.
- 14 How many isomers are there for $\text{C}_3\text{H}_6\text{BrCl}$?

- (A) 3
(B) 4
(C) 5
(D) 6

Use the information provided to answer Questions 15 and 16.

Using 0.100 mol L^{-1} NaOH, a student titrated 25.0 mL of a 0.100 mol L^{-1} weak monoprotic acid, and separately titrated 25.0 mL of a 0.100 mol L^{-1} strong monoprotic acid.

- 15** Which statement about the volume of base required to reach the equivalence point is correct?
- (A) The weak acid will require the same volume of base as the strong acid.
 - (B) The weak acid will require a larger volume of base than the strong acid.
 - (C) The weak acid will require a smaller volume of base than the strong acid.
 - (D) The volume of base required will depend on the molar mass of the acid used.
- 16** Which statement correctly describes the pH at each titration equivalence point?
- (A) The pH of both solutions will be the same.
 - (B) One of the solutions will be neutral while the other will have a pH higher than 7.
 - (C) One of the solutions will be neutral while the other will have a pH lower than 7.
 - (D) One of the solutions will have a pH higher than 7 while the other will have a pH lower than 7.
- 17** The molar heat of combustion of pentan-1-ol is 2800 kJ mol^{-1} . A quantity of pentan-1-ol was combusted, generating 108 kJ of heat.
- What mass of pentan-1-ol was combusted?
- (A) 2.29 g
 - (B) 2.86 g
 - (C) 3.32 g
 - (D) 3.40 g

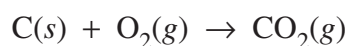
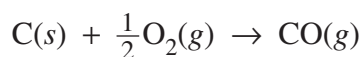
- 18** A household cleaning agent contains a weak base with the formula NaX. 1.00 g of this compound was dissolved in water to give 100.0 mL of solution. A 20.0 mL sample of the solution was titrated with 0.100 mol L⁻¹ hydrochloric acid, and required 24.4 mL of the acid for neutralisation.

What is the molar mass of the weak base?

- (A) 82.0 g
(B) 84.0 g
(C) 122 g
(D) 410 g
- 19** All of the carbon dioxide in a soft drink with an initial mass of 381.04 g was carefully extracted and collected as a gas. The final mass of the drink was 380.41 g.

What volume would the carbon dioxide occupy at 100 kPa and 25°C?

- (A) 0.33 L
(B) 0.35 L
(C) 0.56 L
(D) 0.63 L
- 20** When charcoal reacts in the presence of oxygen, carbon monoxide and carbon dioxide are produced according to the following chemical reactions.



What would be the total mass of gas produced when 400 g of charcoal is reacted, assuming equal amounts are consumed in each reaction?

- (A) 0.93 kg
(B) 1.2 kg
(C) 1.5 kg
(D) 2.5 kg

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Chemistry

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Centre Number

Section I (continued)

Part B – 55 marks

Attempt Questions 21–32

Allow about 1 hour and 40 minutes for this part

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Student Number

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Question 21 (4 marks)

What features of the molecular structure of ethanol account for its extensive use as a solvent? Include a diagram in your answer.

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Question 22 (4 marks)

- (a) Use chemical equations to show how ozone is depleted in the stratosphere. **2**

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- (b) Outline ONE method used to monitor stratospheric ozone. **2**

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Centre Number

Section I – Part B (continued)

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Question 23 (3 marks)

- (a) Element 112 was first synthesised in 1996 and officially named in 2009 as **1**
copernicium, Cn.

Explain why the transuranic isotope $^{278}_{112}\text{Cn}$ is unstable.

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- (b) Describe a method by which transuranic elements can be synthesised. **2**

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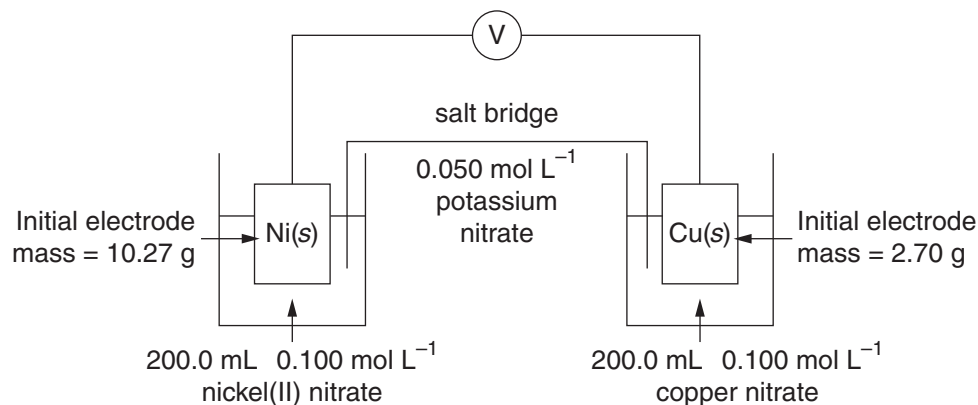
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Question 24 (7 marks)

A galvanic cell was constructed as shown in the diagram.



- (a) Calculate the standard cell potential ($E^\ominus$). In your answer, include a net ionic equation for the overall cell reaction. **2**

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- (b) After a period of time, a solid deposit that had formed on the copper electrode was removed and dried. The deposit had a mass of 0.395 g. **3**

- (i) Calculate the final mass of the nickel electrode.

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- (ii) Calculate the final concentration of the nickel(II) nitrate solution. **2**

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Centre Number

Section I – Part B (continued)

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Question 25 (3 marks)

Explain the role of the conjugate acid/base pair, $\text{H}_2\text{PO}_4^-/\text{HPO}_4^{2-}$, in maintaining the pH of living cells. Include chemical equations in your answer.

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Question 26 (6 marks)

A manufacturer makes lemon cordial by mixing flavouring, sugar syrup and citric acid. The concentration of the citric acid is determined by titration with NaOH.

The sodium hydroxide solution is prepared by dissolving 4.000 g of NaOH pellets in water to give 1.000 L of solution. This solution is standardised by titrating 25.00 mL with a $0.1011 \text{ mol L}^{-1}$ standardised solution of HCl. The average titration volume is found to be 24.10 mL.

To analyse the lemon cordial 50.00 mL of the cordial is diluted to 500.0 mL. Then 25.00 mL of the diluted solution is titrated with the NaOH solution to the phenolphthalein endpoint.

The following data were collected during one of the analysis runs of the lemon cordial.

Titration #1 volume	26.55 mL
Titration #2 volume	27.25 mL
Titration #3 volume	27.30 mL
Titration #4 volume	27.20 mL

- (a) Why is the calculated concentration of the standardised NaOH solution different from the concentration calculated using the mass given, assuming no human error occurred?

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- (b) Determine the concentration of citric acid in the lemon cordial.

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Section I – Part B (continued)

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Student Number

The following extract was taken from a blog about environmental issues.

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Awaiting copyright

Assess the uses of polystyrene and a named biopolymer in terms of their properties, with reference to the statements made in this blog.

[illegible]

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Centre Number

Section I – Part B (continued)

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Student Number

Question 28 (4 marks)

A student investigating the water quality of stormwater in a drain near the school collected samples for testing in the school laboratory. The student conducted the following tests to measure the quality of the stormwater.

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- Hardness
- Phosphate level
- Biochemical oxygen demand
- Total dissolved solids
- Turbidity
- Nitrate level

For TWO of these tests, outline the chemical or physical principle involved and the procedure followed in a school laboratory.

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Question 29 (4 marks)

- (a) Justify the continued use of the Arrhenius definition of acids and bases, despite the development of the more sophisticated Brønsted–Lowry definition. **3**

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- (b) Why does the neutralisation of any strong acid in an aqueous solution by any strong base always result in a heat of reaction of approximately -57 kJ mol^{-1} ? **1**

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Centre Number

Section I – Part B (continued)

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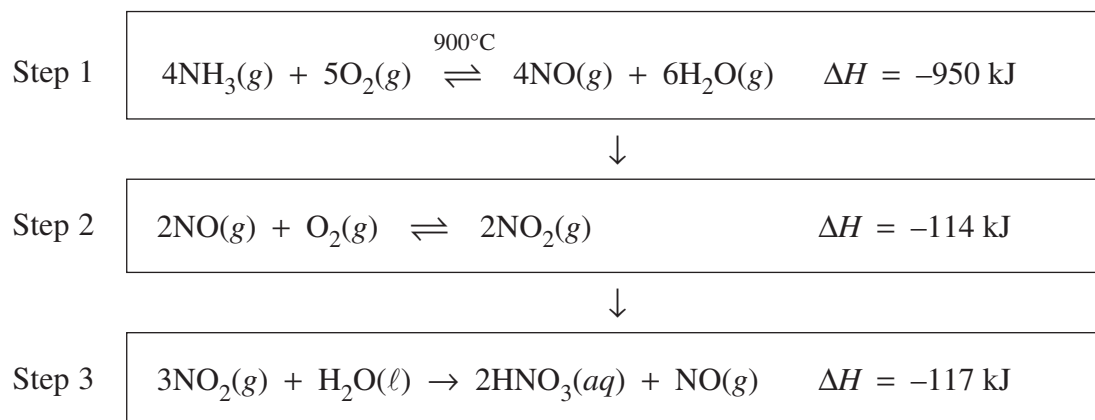
Question 30 (6 marks)

Please turn over

Question 30 (6 marks)

The flowchart outlines the sequence of steps in the Ostwald process for the manufacture of nitric acid.

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Explain the reaction conditions required at each step of the Ostwald process to maximise the yield and production rate of nitric acid.

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Centre Number

Section I – Part B (continued)

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Student Number

Question 31 (4 marks)

A council monitors the water quality of a local river daily. The records for six days are shown in the table. During this period contamination from unknown sources was detected.

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	<i>Mon</i>	<i>Tue</i>	<i>Wed</i>	<i>Thu</i>	<i>Fri</i>	<i>Sat</i>
pH	7.2	7.3	7.2	7.0	7.2	7.3
Turbidity (NTU)	5	67	5	25	10	8
Total dissolved solids (ppm)	340	436	342	380	370	360
Temperature (°C)	15.6	16.0	15.8	15.8	16.0	16.2
Dissolved oxygen (ppm)	9.7	9.6	9.7	4.2	4.3	8.0
Biochemical oxygen demand (ppm)	2.6	4.0	3.2	52	20	7.8
Faecal coliforms (cfu/100 mL)	10	10	12	514	60	18

Propose possible sources of the contamination, justifying your answer with reference to the data provided.

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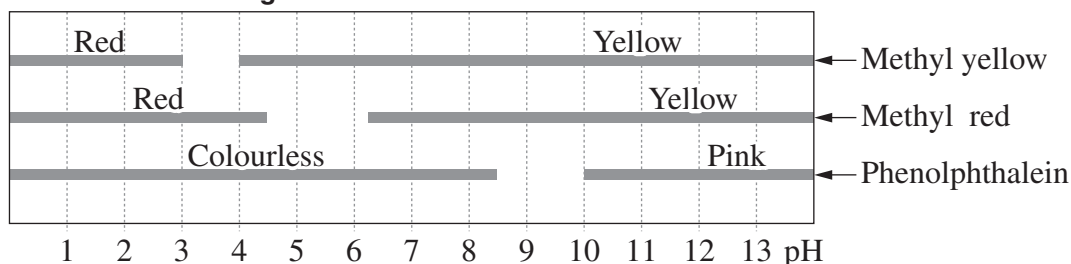
Question 32 (5 marks)

To determine the pH of garden soil, a sample was first saturated with distilled water in a petri dish. Barium sulfate powder was added to the surface of the sample, and drops of the three indicators listed below were added to separate parts of the sample. The colours observed are shown in the table.

Experimental results

<i>Indicator</i>	Methyl yellow	Methyl red	Phenolphthalein
<i>Colour observed</i>	Yellow	Red	Colourless

Indicator colour ranges



Plant response

<i>Plant</i>	<i>soil pH range for optimal growth</i>
Carrot	5.5 – 6.8
Chrysanthemum	6.0 – 6.3
Hydrangea Blue	4.0 – 5.0
Hydrangea White	6.5 – 8.0
Potato	5.0 – 5.7

- (a) Why is barium sulfate powder added when testing soil pH? 1

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- (b) Using the information given, select the plant that will grow well at the current soil pH, and justify your selection. 2

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- (c) Outline the method you would use to test a natural indicator that has been prepared in the school laboratory. 2

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Chemistry

Section II

25 marks

Attempt ONE question from Questions 33–37

Allow about 45 minutes for this section

Answer parts (a)–(c) of the question in Section II Answer Booklet 1.

Answer parts (d)–(e) of the question in Section II Answer Booklet 2.

Extra writing booklets are available.

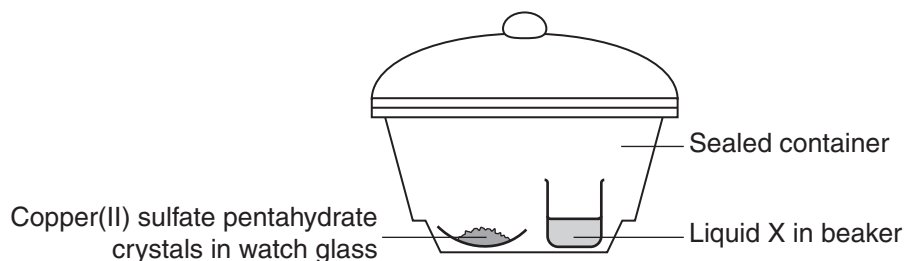
Show all relevant working in questions involving calculations.

	Pages
Question 33 Industrial Chemistry	24–25
Question 34 Shipwrecks, Corrosion and Conservation	26–27
Question 35 The Biochemistry of Movement	28–29
Question 36 The Chemistry of Art	30–31
Question 37 Forensic Chemistry	32–33

Question 33 — Industrial Chemistry (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) A student set up the following experiment to model a process and observed the colour change of the crystals. 3



Identify liquid X and explain the colour change of the crystals. Include a chemical equation in your answer.

- (b) (i) Why are the large volumes of $\text{CO}_2(\text{g})$ produced during the Solvay process of little environmental concern? Include chemical equations in your answer. 3
- (ii) Calcium chloride is also produced during the Solvay process. Calculate the mass of calcium chloride produced per tonne of sodium chloride used in the Solvay process. 2
- (c) A 0.05 mol L^{-1} solution of sodium chloride was electrolysed using graphite electrodes. Separate pieces of litmus paper were dipped into the solution next to each electrode.

The following observations were made.

<i>Polarity of electrode</i>	<i>Observation</i>	<i>Colour of litmus paper</i>
Positive	Bubbles	Red
Negative	Bubbles	Blue

- (i) Account for the observations at the anode and cathode. Include relevant chemical equations in your answer. 3
- (ii) What is the difference between the electrolytic cell described above and a galvanic cell, in terms of energy requirements? 2

Question 33 continues on page 25

Question 33 (continued)

Answer parts (d)–(e) in Section II Answer Booklet 2.

- | | | | |
|-----|------|---|----------|
| (d) | (i) | Models are often used to help explain complex concepts. Outline a first-hand investigation that can model an equilibrium reaction. | 2 |
| | (ii) | Assess the validity of the information that could be collected in this investigation. | 3 |
| (e) | | Evaluate the impact on society of the environmental issues associated with THREE of the industrial processes that you have studied in this option. | 7 |

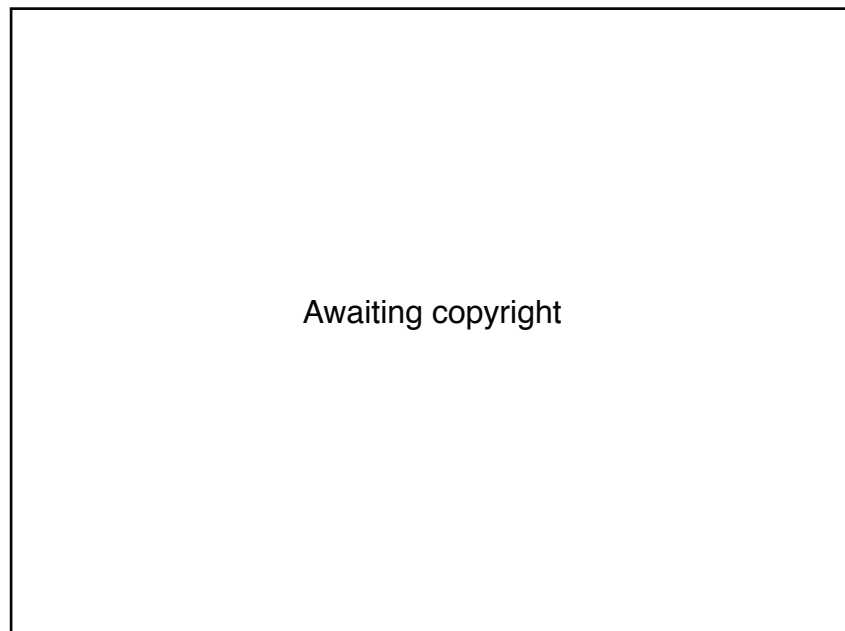
End of Question 33

Question 34 — Shipwrecks, Corrosion and Conservation (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) The diagram shows a hot water tank.

3



Name an appropriate material from which X can be made, and justify your choice.

- (b) There are two common household methods to remove the tarnish on silver cutlery: polishing with a soft cloth and toothpaste or placing the cutlery into an aluminium tray filled with warm sodium hydrogen carbonate solution and leaving it overnight.

- (i) Explain, with the use of equations, the chemistry involved in the aluminium tray method.

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- (ii) Identify an advantage of the aluminium tray method.

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Question 34 continues on page 27

Question 34 (continued)

- (c) A dilute solution of potassium sulfate was electrolysed using graphite electrodes. Separate pieces of litmus paper were dipped into the solution next to each electrode.

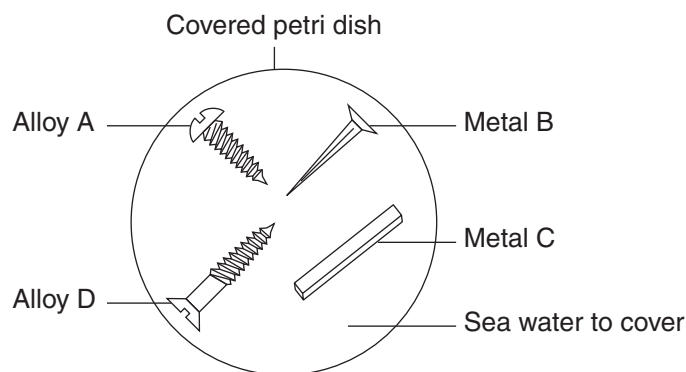
The following observations were made.

<i>Polarity of electrode</i>	<i>Observation</i>	<i>Colour of litmus paper</i>
Positive	Bubbles	Red
Negative	Bubbles	Blue

- (i) Draw and label a diagram to represent this cell. 3
- (ii) Identify the products formed at the anode and cathode by writing the equations for each of these reactions. 2

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) A student set up the following investigation to compare the rate of corrosion of a variety of metals and alloys, in order to identify those best suited for use in marine vessels. Photos were taken of the experiment on a daily basis for several weeks.



- (i) Explain the method the student should have used to prepare the samples for the investigation. 2
- (ii) Assess the validity of the data collected in this experiment. 3
- (e) ‘Corrosion at great depth in the ocean was predicted to be slow.’ 7

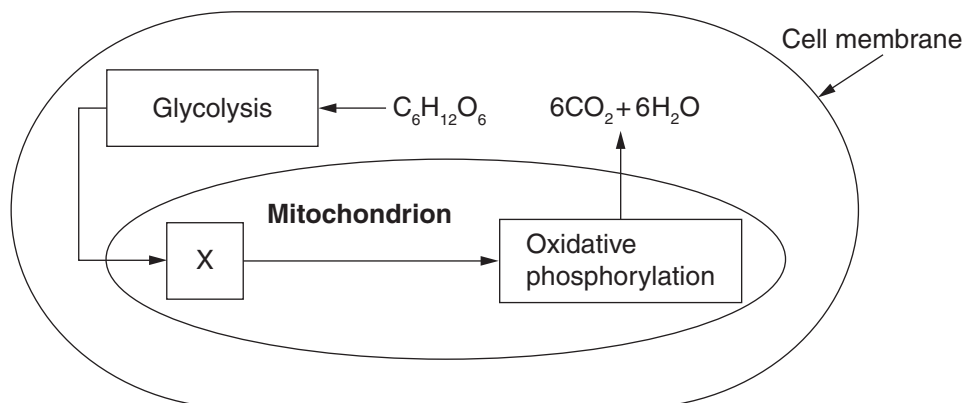
Evaluate this statement with reference to recent maritime investigations. Include chemical equations in your answer.

End of Question 34

Question 35 — The Biochemistry of Movement (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) The main steps in cellular respiration are shown in the diagram of the cell. 3



Identify the process represented by X, and explain the difference in oxygen requirements of glycolysis and the chemical processes in mitochondria.

- (b) (i) Using the general formula for an amino acid, write an equation to show the formation of a dipeptide. 2
- (ii) Explain the effect of an increase in temperature and a change in pH on the molecular structure of a protein. 3

- (c) The table shows the percentage of each muscle cell type used for particular activities.

Activity	% of Type 1 muscle cells	% of Type 2 muscle cells
Sedentary activity	40	15
Sprinting	25	50
Marathon running	75	10

- (i) The use of muscle cell type changes as the nature of the activity changes. Give reasons for this. 3
- (ii) Use an equation to show how ATP is regenerated during muscle contraction. 2

Question 35 continues on page 29

Question 35 (continued)

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) A student wants to test the effects of temperature on enzyme function. A piece of calf liver is placed in a test tube with hydrogen peroxide. The test tube is then placed in a water bath. As the reaction occurs, the rate is monitored by measuring the time taken for the reaction to stop producing oxygen.
- (i) Account for the fact that there is an optimal temperature for enzyme function. **2**
- (ii) Assess the validity of the data collected in this experiment. **3**
- (e) ‘A knowledge of the availability of energy from fats and carbohydrates has improved society’s understanding of nutrition.’ **7**

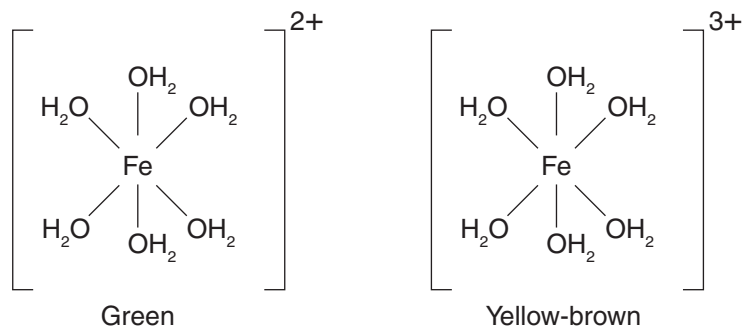
Analyse this statement with reference to metabolic pathways.

End of Question 35

Question 36 — The Chemistry of Art (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) The structures of two aqueous iron complexes are represented in the diagram. 3



Account for the different colours of the complexes.

- (b) (i) Identify ONE cation and ONE anion that can be represented by the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6$. 2
- (ii) Explain the fact that Group I and Group II metal ions have one oxidation state while the transition metals often have multiple oxidation states. 3

- (c) The emission lines for an element are listed in the table.

<i>Line number</i>	<i>Wavelength (nm)</i>
1	656
2	486
3	434
4	419

- (i) Describe a method by which the values in the table may have been obtained. 2
- (ii) Draw an energy level diagram that corresponds to the data given. Assume a common final energy level for each transition. 3

Question 36 continues on page 31

Question 36 (continued)

Answer parts (d)–(e) in Section II Answer Booklet 2.

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|-----|------|---|----------|
| (d) | (i) | Describe an experiment that can be performed in the school laboratory to illustrate the emission colour of Group II metals. | 2 |
| | (ii) | Assess the validity of the data collected in this experiment. | 3 |
| (e) | | To what extent has our understanding of the pigments used in medieval painting increased with the use of a range of technologies? | 7 |

In your answer, refer to **THREE** spectroscopic techniques.

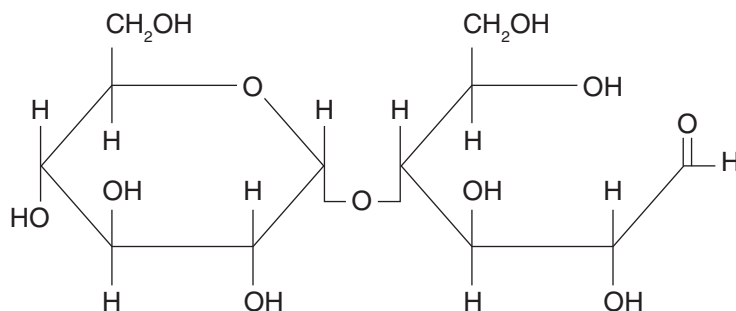
End of Question 36

Question 37 — Forensic Chemistry (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) Identify the structure shown, and justify your answer.

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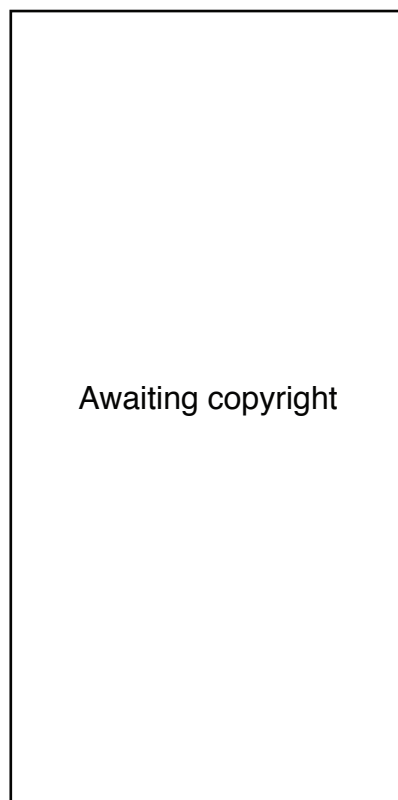
- (b) (i) Using the general formula of an amino acid, write an equation to show the formation of a dipeptide. **2**
- (ii) Explain the principles of paper chromatography, with reference to the separation of amino acids. **3**

Question 37 continues on page 33

Question 37 (continued)

- (c) (i) Describe the technique by which DNA is analysed and then used to identify relationships between people. **4**
- (ii) The diagram shows the DNA fingerprints (W, X, Y and Z) from a child, the child's biological parents and a family friend. **1**

Identify the DNA fingerprint of the child.



Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) A student has four unlabelled bottles all containing white crystals. The four substances are known to be sucrose, potassium chloride, glycine (amino acid) and acetylsalicylic acid (aspirin).
- (i) Explain a method the student could use in the school laboratory to identify the contents of the four unlabelled bottles. **4**
- (ii) Identify ONE destructive and ONE non-destructive test the student may have used. **1**
- (e) Select THREE instrumental techniques you have studied in this option. Evaluate the optimal use of each technique with reference to the analysis of forensic samples. **7**

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DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY																2 He 4.003 Helium																	
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number Symbol Standard Atomic Weight Name																5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine		10 Ne 20.18 Neon					
		12 Mg 24.31 Magnesium																		13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine		18 Ar 39.95 Argon					
19 K 39.10 Potassium		20 Ca 40.08 Calcium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.38 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton	
37 Rb 85.47 Rubidium		38 Sr 87.61 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.96 Molybdenum		43 Tc Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon	
55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57–71 Lanthanoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.9 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po Polonium		85 At Astatine		86 Rn Radon	
87 Fr Francium		88 Ra Radium		89–103 Actinoids		104 Rf Rutherfordium		105 Db Dubnium		106 Sg Seaborgium		107 Bh Bohrium		108 Hs Hassium		109 Mt Meitnerium		110 Ds Darmstadtium		111 Rg Roentgenium		112 Cn Copernicium													

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

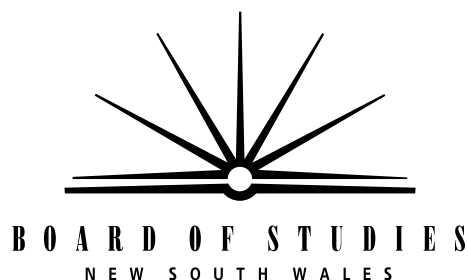
89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
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Elements with atomic numbers 113 and above have been reported but not fully authenticated.

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of data. Some data may have been modified.



2011
HIGHER SCHOOL CERTIFICATE
EXAMINATION

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Centre Number

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Student Number

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Question Number

Chemistry

Section II Answer Booklet 1

Question 33	Industrial Chemistry	Parts (a), (b) and (c)
Question 34	Shipwrecks, Corrosion and Conservation	Parts (a), (b) and (c)
Question 35	The Biochemistry of Movement	Parts (a), (b) and (c)
Question 36	The Chemistry of Art	Parts (a), (b) and (c)
Question 37	Forensic Chemistry	Parts (a), (b) and (c)

Instructions

- Answer ONE question from Questions 33–37 in this booklet and in the Section II Answer Booklet 2
- Write your Centre Number and Student Number at the top of this page
- Write the question number in the space provided

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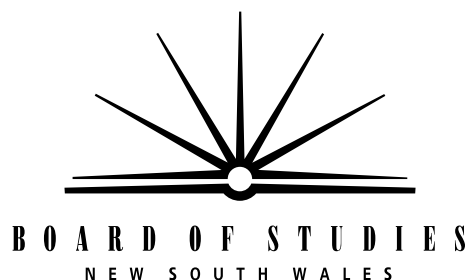
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Student Number

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Question Number

Chemistry

Section II Answer Booklet 2

- Question 33 Industrial Chemistry Parts (d) and (e)
- Question 34 Shipwrecks, Corrosion and Conservation Parts (d) and (e)
- Question 35 The Biochemistry of Movement Parts (d) and (e)
- Question 36 The Chemistry of Art Parts (d) and (e)
- Question 37 Forensic Chemistry Parts (d) and (e)

Instructions

- Answer ONE question from Questions 33–37 in this booklet and in the Section II Answer Booklet 1
- Write your Centre Number and Student Number at the top of this page
- Write the question number in the space provided

(d)

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2011 HSC Chemistry Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	A
2	D
3	C
4	D
5	B
6	A
7	B
8	C
9	A
10	C
11	D
12	D
13	C
14	C
15	A
16	B
17	D
18	A
19	B
20	B

Section I, Part B

Question 21

Criteria	Marks
- Demonstrates a thorough knowledge and understanding of the molecular structure and use of ethanol as a solvent - Provides an appropriate diagram, eg ethanol hydrogen bonding to water - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	4
- Demonstrates a sound knowledge and understanding of the structure and use of ethanol as a solvent - Provides correct structure for ethanol - Communicates some scientific principles and ideas in a clear manner	3
Demonstrates a basic knowledge of the structure and use of ethanol as a solvent	2
- Demonstrates a limited knowledge of the structure and use of ethanol as a solvent OR - Provides correct structure for ethanol	1

Question 22 (a)

Criteria	Marks
Shows full depletion of ozone with correct equations	2
Gives ONE correct equation	1

Question 22 (b)

Criteria	Marks
Sketches in general terms a method used for monitoring ozone	2
States a method used	1

Question 23 (a)

Criteria	Marks
Gives ONE correct reason why an isotope such as copernicium-278 is unstable	1

Question 23 (b)

Criteria	Marks
• Provides features of a method by which transuranic elements can be synthesised	2
• Identifies a correct method by which transuranic elements can be synthesised	1

Question 24 (a)

Criteria	Marks
• Calculates correct $E^\ominus$ • Provides correctly balanced net ionic equation	2
• Calculates correct $E^\ominus$ OR • Provides correctly balanced net ionic equation	1

Question 24 (b) (i)

Criteria	Marks
• Correctly calculates final mass of Ni electrode	3
• Correctly calculates moles of Cu deposited • Calculates mass of Ni lost	2
• Correctly calculates moles of Cu deposited	1

Question 24 (b) (ii)

Criteria	Marks
• Correctly calculates final concentration of Ni solution using moles of Ni lost from b (i)	2
• Includes one correct relevant mole calculation	1

Question 25

Criteria	Marks
• States the role of $\text{H}_2\text{PO}_4^-/\text{HPO}_4^{2-}$ as a buffer system in the cell • Provides TWO correct equations	3
• States the role of $\text{H}_2\text{PO}_4^-/\text{HPO}_4^{2-}$ as a buffer system in the cell AND provides ONE correct equation OR • Provides TWO correct equations	2
• States the role of $\text{H}_2\text{PO}_4^-/\text{HPO}_4^{2-}$ as a buffer system in the cell OR • Provides ONE correct equation	1

Question 26 (a)

Criteria	Marks
- Provides a valid reason - Correctly calculates both NaOH concentrations OR - Indicates actual value is less than theoretical value	2
- Provides a valid reason OR - Correctly calculates TWO NaOH concentrations	1

Question 26 (b)

Criteria	Marks
Correctly calculates concentration of original solution of citric acid	4
- Calculates concentration of original solution of citric acid using incorrect NaOH concentration or incorrect mole ratio or incorrect volume of HCl OR - Correctly calculates concentration of diluted solution of citric acid	3
Calculates concentration of citric acid solution with TWO errors	2
Provides ONE correct step in the calculation	1

Question 27

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of the uses of BOTH polymers in terms of their properties • Refers to polystyrene AND a named biopolymer • Makes a judgement of the extract based on criteria • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	5
• Demonstrates a sound knowledge and understanding of the uses of BOTH polymers in terms of their properties • Refers to polystyrene AND a named biopolymer • Makes a judgement of the extract based on criteria • Communicates scientific principles and ideas in a clear manner	4
• Demonstrates basic knowledge and understanding of the use of BOTH polymers in terms of their properties • Refers to polystyrene AND a named biopolymer • Communicates some scientific principles and ideas in a clear manner	3
• Demonstrates limited knowledge and understanding of the use of polymers in terms of their properties • Refers to polystyrene only or a named biopolymer • Communicates ideas in a basic form using general scientific terms	2
• Demonstrates some knowledge and understanding of the uses of polymers or their properties • Communicates simple ideas	1

Question 28

Criteria	Marks
- Demonstrates a thorough knowledge and understanding of TWO tests - Provides the main features of the chemical or physical principle of EACH test - Provides the main features of the procedure used for EACH test	4
- Demonstrates a sound knowledge and understanding of TWO tests - Outlines the chemical or physical principle of EACH test AND the procedure used for ONE test OR - Outlines the procedure used for EACH test AND the chemical or physical principle of ONE test	3
- Demonstrates a basic knowledge and understanding of at least ONE test - Outlines the principle of EACH test OR - Outlines the procedure used for EACH test OR - Outlines the principle AND procedure used for ONE test only	2
- Demonstrates a limited knowledge and understanding of ONE test - Outlines the principle of ONE test OR - Outlines the procedure of ONE test	1

Question 29 (a)

Criteria	Marks
- Demonstrates thorough knowledge and understanding of the Arrhenius AND Brönsted–Lowry definitions of acids and bases - Supports the statement with evidence	3
- Demonstrates a sound knowledge and understanding of the Arrhenius AND Brönsted–Lowry definitions OR - Demonstrates a sound knowledge and understanding of EITHER the Brönsted–Lowry OR the Arrhenius definition - Supports the statement with evidence OR - Demonstrates a basic knowledge and understanding of the Arrhenius AND Brönsted–Lowry definitions - Supports the statement with evidence	2
- Demonstrates a sound knowledge and understanding of EITHER the Arrhenius OR the Brönsted–Lowry definition OR - Provides a justification	1

Question 29 (b)

Criteria	Marks
Provides a reason why the heat of reaction is approximately -57 kJ mol^{-1} for neutralisation reactions of strong acids and strong bases	1

Question 30

Criteria	Marks
- Demonstrates a thorough knowledge and understanding of equilibrium and Le Chateliers principle with reference to the THREE reactions - Relates the conditions required to the increased yield and production rate - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6
- Demonstrates a sound knowledge and understanding of equilibrium and Le Chateliers principle with reference to the THREE reactions - Communicates some scientific principles and ideas in a clear manner	4–5
- Demonstrates a basic knowledge and understanding of equilibrium and Le Chateliers principle with reference to flow chart - Communicates ideas in a basic form using general scientific language	2–3
- Demonstrates a limited knowledge and understanding of equilibrium and Le Chateliers principle - Communicates simple ideas	1

Question 31

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of potential sources of contamination of waterways • Provides evidence of a thorough analysis of the data • Provides reasons for TWO possible sources of contamination • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	4
• Demonstrates a sound knowledge and understanding of potential sources of contamination of waterways • Refers to data • Provides reasons for TWO possible sources of contamination • Communicates some scientific principles and ideas in a clear manner	3
• Demonstrates a basic knowledge and understanding of potential sources of contamination of waterways • Refers to data • Provides reasons for ONE possible source of contamination OR states TWO possible sources of contamination • Communicates ideas in a basic form using general scientific terms	2
• Demonstrates a limited knowledge of the potential sources of contamination of waterways • Communicates simple ideas • States at least ONE source of contamination	1

Question 32 (a)

Criteria	Marks
• Provides correct reason	1

Question 32 (b)

Criteria	Marks
• Determines correct soil pH resulting in identification of suitable plant	2
• Identifies correct plant OR • Determines correct soil pH	1

Question 32 (c)

Criteria	Marks
• Sketches in general terms a method using several solutions of known pH	2
• Sketches in general terms a method OR • Provides some solutions to be tested	1

Section II

Question 33 (a)

Criteria	Marks
- Relates cause and effect of the colour change - Provides a relevant chemical equation	3
Outlines relevant features of the chemical process involved	2
Provides a relevant feature of the process involved	1

Question 33 (b) (i)

Criteria	Marks
- Provides chemical equations showing production and usage of $\text{CO}_2(g)$ in the Solvay process - States a reason to support statement	3
- Provides a chemical equation demonstrating either production or usage of $\text{CO}_2(g)$ in the Solvay process - States a reason to support statement	2
- Provides a chemical equation demonstrating production or usage of $\text{CO}_2(g)$ in the Solvay process OR - States a reason to support statement	1

Question 33 (b) (ii)

Criteria	Marks
Calculates correct mass of CaCl_2	2
- Uses correct molar ratio - Uses correct moles NaCl OR - Uses correct method with incorrect molar ratio	1

Question 33 (c) (i)

Criteria	Marks
- Provides reasons for ALL observations - Provides BOTH fully balanced chemical half equations	3
- Provides reasons for SOME observations - Provides BOTH fully balanced chemical half equations OR - Provides reasons for ALL observations - Provides ONE fully balanced chemical half equation	2
- Provides reasons for SOME observations OR - Provides ONE fully balanced chemical equation	1

Question 33 (c) (ii)

Criteria	Marks
Provides reasons for TWO specific energy conversions and their requirements	2
Provides a general statement regarding energy requirements	1

Question 33 (d) (i)

Criteria	Marks
Indicates the main features of a physical model that demonstrates equilibrium	2
States a feature of or sketch of a physical procedure that models equilibrium	1

Question 33 (d) (ii)

Criteria	Marks
- Gives a value judgement based upon the information collected providing factors that promote validity - Indicates the limitations of the model that compromise validity	3
- Discusses factors that promote validity of the information collected - Gives a value judgement	2
- Identifies a limitation that relates to the model that compromises its validity OR - Provides a factor that relates to the model that promotes validity	1

Question 33 (e)

Criteria	Marks
- Demonstrates thorough knowledge and understanding of the environmental issues of THREE industrial processes studied - Provides a judgement about these issues in terms of the overall impact on society - Uses correct and relevant chemistry - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
- Demonstrates sound knowledge and understanding of the environmental issues of THREE industrial processes studied - Outlines the overall impact that these issues have had on society - Uses some correct and relevant chemistry - Communicates some scientific principles and ideas in a clear manner	4–5
- Demonstrates a basic understanding of the environmental issues of the industrial processes studied - Identifies some impacts these issues have had on society - Communicates ideas in a basic form using general scientific terms	2–3
- Demonstrates a limited knowledge and understanding of the environmental issues of the industrial processes studied - Communicates simple ideas	1

Question 34 (a)

Criteria	Marks
- Identifies a suitable material for X - States that the sacrificial anode, X, is preferentially oxidised - Relates this to the relevant reduction potentials for X and iron or relative activity	3
- Identifies that corrosion is prevented by preferential oxidation of the named sacrificial anode OR - States relevant reduction potentials for an appropriate X, and iron or relative activity	2
- States that X acts as a sacrificial anode OR - Identifies a suitable material for X OR - States that X is oxidised	1

Question 34 (b) (i)

Criteria	Marks
- Provides an overall balanced equation for the relevant cell - Clearly identifies the structure of this galvanic cell including electrolyte	4
- Provides correct half equations for this cell - States that aluminium is the more reactive metal - States that a galvanic cell is formed	3
- TWO of the following: - States that aluminium is more reactive than silver - States that aluminium is oxidised - States that a galvanic cell is formed	2
- States that aluminium is more reactive than silver OR - States that aluminium is oxidised OR - States that a galvanic cell is formed	1

Question 34 (b) (ii)

Criteria	Marks
States that the electrochemical method is restorative, and hence preferable	1

Question 34 (c) (i)

Criteria	Marks
Provides a detailed diagram of the cell	3
Provides a basic diagram of the cell	2
Provides a limited diagram of the cell	1

Question 34 (c) (ii)

Criteria	Marks
Provides balanced equations at correct electrodes	2
- Provides ONE balanced equation at correct electrode OR - Provides TWO balanced equations but incorrect electrodes	1

Question 34 (d) (i)

Criteria	Marks
- Provides features of an appropriate method - Provides a reason for the treatment	2
- Provides features of an appropriate method OR - Provides a reason for the treatment	1

Question 34 (d) (ii)

Criteria	Marks
Provides a value judgement on the validity of the data collected with examples of limitations and positive attributes of the experimental design	3
- Discusses the limitations and positive attributes of experimental design OR - Discusses limitations or attributes of experimental design and gives a judgement	2
- Gives value judgement in terms of validity of the data collected OR - Identifies a limitation of the investigation's design OR - Identifies an attribute OR - Identifies a controlled variable OR - Relates validity to control of variables	1

Question 34 (e)

Criteria	Marks
- Demonstrates thorough knowledge and understanding of the significance of the conditions prevailing at great depths in terms of corrosion of shipwrecks - Makes value judgement based on criteria (or evidence) - Uses correct and relevant chemistry with correct equations - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
- Demonstrates sound knowledge and understanding of the significance of the conditions prevailing at great depths in terms of corrosion of shipwrecks - Writes a relevant and correct equation - Communicates some scientific principles and ideas clearly	4–5
- Demonstrates a basic knowledge of the significance of the conditions prevailing at great depths and their effect on corrosion of shipwrecks - Communicates ideas in a basic form using general scientific terms	2–3
- Demonstrates a limited knowledge of conditions at great depths and their effect on corrosion of a shipwreck - Communicates simple ideas	1

Question 35 (a)

Criteria	Marks
- Correctly identifies process X - Shows the difference in oxygen requirements by identifying glycolysis as anaerobic and the chemical processes in the mitochondria as aerobic	3
- Correctly identifies process X - Identifies glycolysis as anaerobic OR - Correctly identifies process X - Identifies the chemical processes in the mitochondria as aerobic OR - Shows the difference in oxygen requirements by identifying glycolysis as anaerobic and the chemical processes in the mitochondria	2
- Correctly identifies process X OR - Identifies glycolysis as anaerobic OR - Identifies the chemical processes in the mitochondria as aerobic	1

Question 35 (b) (i)

Criteria	Marks
• Provides correctly balanced equation	2
• Provides the general formula of an amino acid	1

Question 35 (b) (ii)

Criteria	Marks
- Clearly explains how changes in pH and increases in temperature will affect the molecular structure of a protein - Refers to secondary and tertiary structure	3
- States how changes in pH will affect the molecular structure of a protein - States how increases in temperature will affect the molecular structure of a protein	2
- States how changes in pH will affect the molecular structure of a protein OR - States how increases in temperature will affect the molecular structure of a protein	1

Question 35 (c) (i)

Criteria	Marks
- Demonstrates a thorough knowledge and understanding of Types 1 and 2 muscle cells - Provides reasons for the different usage of muscle cell types - Refers to EACH activity - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	3
- Demonstrates a sound knowledge and understanding of Types 1 and 2 muscle cells - Refers to at least TWO activities - Communicates some scientific principles and ideas in a clear manner	2
- Demonstrates a basic knowledge and understanding of Types 1 and 2 muscle cells - Refers to at least ONE activity - Communicates ideas in a basic form using general scientific terms	1

Question 35 (c) (ii)

Criteria	Marks
Demonstrates a thorough knowledge of the chemistry involved by using a relevant chemical equation showing the production of ATP including consideration of energy	2
- Demonstrates a knowledge of some of the chemistry involved by using a relevant chemical equation showing the production of ATP without the addition of energy OR - Demonstrates a knowledge of some of the chemistry involved by using a chemical equation to show the formation of ADP (Energy $\text{ATP} \rightarrow \text{ADP} + \text{Pi}$)	1

Question 35 (d) (i)

Criteria	Marks
Provides reasons for the fact that there is an optimal temperature for enzyme function	2
- Provides only a simple reason for an optimal temperature OR - Gives a basic description of the effect of temperature on enzyme function	1

Question 35 (d) (ii)

Criteria	Marks
Gives a value judgement on the validity of the data collected with examples of limitations and positive attributes of the experimental design	3
- Discusses the interpretation of the independent variable in this investigation OR - Discusses the control of variables in this investigation OR - Analyses the way results are collected in this investigation	2
- Gives a value judgement in terms of the validity of the investigation OR - Identifies a limitation of the investigation's validity OR - Relates validity to control of variables OR - Identifies a controlled variable	1

Question 35 (e)

Criteria	Marks
- Demonstrates a thorough knowledge and understanding of the energy availability of fats and carbohydrates - Explains how this knowledge has improved society's understanding of nutrition - Refers to metabolic pathways - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7
- Demonstrates a sound knowledge and understanding of the energy availability of fats and carbohydrates - Describes how this knowledge has improved society's understanding of nutrition - Refers to at least ONE metabolic pathway - Communicates some scientific principles and ideas in a clear manner	5–6
- Demonstrates a basic knowledge and understanding of the energy availability of fats and carbohydrates OR - Describes how this knowledge has improved society's understanding of nutrition - Refers to at least ONE metabolic pathway - Communicates ideas in a basic form using general scientific terms	3–4
- Demonstrates a limited knowledge and understanding of the energy availability of fats and carbohydrates - Communicates simple ideas	1–2

Question 36 (a)

Criteria	Marks
Provides detailed reasons for the difference in colour	3
Provides reasons for the difference in colour	2
Provides a reason for the difference in colour	1

Question 36 (b) (i)

Criteria	Marks
Provides one correct cation and one correct anion	2
Provides one correct cation or anion	1

Question 36 (b) (ii)

Criteria	Marks
- Identifies THREE key points of electron configuration, orbital energy and multiple electron loss in transition elements - Clearly shows the relationship between the properties of Groups I and II metals and the transition metals	3
Identifies TWO key points of electron configuration, orbital energy and multiple electron loss in transition elements	2
Identifies ONE of the key points of electron configuration, orbital energy and multiple electron loss in transition elements	1

Question 36 (c) (i)

Criteria	Marks
Provides features of the method to obtain values in the table	2
Includes one of: sample emission or measurement of emitted radiation	1

Question 36 (c) (ii)

Criteria	Marks
Provides a diagram showing FOUR transition arrows, all with the same final energy level, with arrow heads showing downward direction and labelled with correct wavelengths with appropriate spacing	3
Provides a diagram showing THREE correct features such as arrow directions, common final energy level, correct labelling for at least TWO transitions	2
Provides a diagram showing TWO correct features such as arrow directions, common final energy level, correct labelling for at least ONE transition	1

Question 36 (d) (i)

Criteria	Marks
Provides features of the experiment	2
Provides a limited feature of the experiment	1

Question 36 (d) (ii)

Criteria	Marks
Gives a value judgement on the validity of the data collected with examples of limitations and positive attributes of the experimental design	3
- Discusses limitations and positive attributes of experimental design OR - Discusses limitations or attributes of experimental design and gives a judgement of the validity of the data	2
- Gives a value judgement in terms of the validity of the investigation OR - Identifies a limitation or a positive attribute of the investigation's validity OR - Identifies a controlled variable	1

Question 36 (e)

Criteria	Marks
- Demonstrates a thorough knowledge and understanding of the techniques and the composition of pigments, AND relates this to spectroscopic techniques - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	7
- Demonstrates a sound knowledge and understanding of the techniques and the composition of pigments AND relates this to spectroscopic techniques - Communicates some scientific principles and ideas clearly	5–6
- Demonstrates a basic knowledge and understanding of the techniques OR the composition of pigments - Communicates ideas in a basic form using general scientific terms	3–4
- Demonstrates a limited knowledge of the technique OR pigments - Communicates simple ideas	1–2

Question 37 (a)

Criteria	Marks
- Correctly names the structure - Identifies the free carbonyl group - States that it can be oxidised/reducing sugar	3
- Correctly names the structure - Identifies the free carbonyl group OR - States that it can be oxidised/reducing sugar	2
- Names the structure as a sugar OR - Identifies the free carbonyl group	1

Question 37 (b) (i)

Criteria	Marks
Provides correctly balanced equation	2
Provides formula for amino acid	1

Question 37 (b) (ii)

Criteria	Marks
Shows clearly how the principles of paper chromatography can separate to amino acids	3
Describes the principles of paper chromatography with reference to amino acids	2
Demonstrates a basic knowledge of the principles of paper chromatography	1

Question 37 (c) (i)

Criteria	Marks
- Demonstrates a thorough knowledge of the principle of the analysis of DNA - States how it is used in the identification of relationships	4
- Demonstrates a sound knowledge of the principle of the analysis of DNA - States how it is used in the identification of relationships	3
Demonstrates a basic knowledge of the principle of the analysis of DNA and how it can be used to identify relationships	2
- Demonstrates a limited knowledge of the principle of the analysis of DNA OR - States how it can be used to identify relationships	1

Question 37 (c) (ii)

Criteria	Marks
• Correctly identifies the DNA of the child	1

Question 37 (d) (i)

Criteria	Marks
• Demonstrates a thorough knowledge of the principles of identification of all FOUR unknown compounds • Relates the tests to the properties of the compounds • Demonstrates coherence and logical progression of ideas	4
• Demonstrates a sound knowledge of the principles of identification of THREE unknown compounds • Communicates ideas in a clear manner	3
• Demonstrates a basic knowledge of the principles of identification of the unknown compounds • Communicates ideas in a basic form	2
• Demonstrates a limited knowledge of the principles of identification of the unknown compounds • Communicates simple ideas	1

Question 37 (d) (ii)

Criteria	Marks
• Correctly identifies both tests	1

Question 37 (e)

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of THREE instrumental techniques and their optimal use in forensic analysis • Makes a value judgement for each instrument • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates a sound knowledge and understanding of THREE instrumental techniques and their optimal use in forensic analysis • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a basic knowledge and understanding of the instrumental techniques and their use in forensic analysis • Communicates in a basic form using general scientific terms	2–3
• Demonstrates a limited knowledge and understanding of the instrumental techniques and their use • Communicates simple ideas	1

Chemistry

2011 HSC Examination Mapping Grid

Section I Part A

Question	Marks	Content	Syllabus outcomes
1	1	9.2.3.2.2, 9.2.3.3.1	H9
2	1	9.2.3.2.5/6, 9.2.3.3.4/5	H9
3	1	9.3.1.2.1, 9.3.1.3.3	H8
4	1	9.4.4.2.8, 9.4.4.3.1	H8, H9
5	1	9.4.5.2.3, 9.4.5.3.3	H8
6	1	9.2.3.2.9, 9.3.5.2.3	H9
7	1	9.2.5.2.1	H6
8	1	9.3.5.2.2	H9
9	1	9.4.4.2.6	H6
10	1	9.4.3.2.1, 9.4.3.3.1	H8, H14
11	1	9.2.1.2.3, 9.2.1.3.2, 9.4.4.2.9	H9, H10
12	1	9.2.4.2.2, 9.2.4.3.4	H7, H8, H14
13	1	9.3.2.2.4, 9.3.2.2.5	H8, H14
14	1	9.4.4.2.9	H13
15	1	9.3.3.2.4, 9.3.3.3.4, 9.3.4.3.3	H8, H10, H14
16	1	9.3.3.2.3, 9.3.4.2.4, 9.3.4.3.2	H8, H14
17	1	9.2.3.2.7/9	H9, H12, H13
18	1	9.3.4.3.3	H8, H10, H12
19	1	9.3.2.2.9, 9.3.2.3.1	H12
20	1	9.4.1.2.3	H9, H10, H12

Section I Part B

Question	Marks	Content	Syllabus outcomes
21	4	9.2.3.2.3, 9.2.3.2.9	H9, H13, H14
22 (a)	2	9.4.4.3.1	H6, H9, H13
22 (b)	2	9.4.4.2.11	H6, H11
23 (a)	1	9.2.5.2.1, 9.2.5.2.2	H6, H14
23 (b)	2	9.2.5.2.2	H6, H12
24 (a)	2	9.2.4.3.4, 9.2.4.2.4	H7, H12, H13
24 (b) (i)	3	9.2.4.2.1, 9.2.4.2.4	H10, H12, H14
24 (b) (ii)	2	9.2.4.2.1, 9.2.4.2.4	H10, H12, H14
25	3	9.3.4.2.3/5/6/9	H8, H10, H13
26 (a)	2	9.3.4.3.3, 9.3.4.2.8	H7, H10, H12, H14
26 (b)	4	9.3.4.3.3, 9.3.4.2.8	H10, H12, H14
27	5	9.2.2.3.1, 9.2.1.2.8	H1, H4, H9, H12, H14
28	4	9.4.5.2.1, 9.4.5.3.1/2, 9.4.3.3.1	H8, H11, H14

29 (a)	3	9.3.4.2.1, 9.3.4.2.2	H2, H8, H14
29 (b)	1	9.3.4.2.7	H7, H14
30	6	9.3.2.2.4, 9.4.2.2.5/6/7/8	H7, H8, H10, H14
31	4	9.4.5.2.1/2, 9.4.5.3.3	H14
32 (a)	1	9.3.1.2.3, 9.4.3.3.3, 9.4.3.3.1	H6, H14
32 (b)	2	9.3.1.2.2/3	H4, H14
32 (c)	2	9.3.1.2.2, 9.3.1.3.1	H8, H14

Section II

Question 33 — Industrial Chemistry

Question	Marks	Content	Syllabus outcomes
(a)	3	9.5.3.2.6	H7, H8, H13
(b) (i)	3	9.5.6.2.3, 9.5.6.2.4	H4, H8, H13, H14
(b) (ii)	2	9.5.6.2.3, 9.5.6.3.2	H8, H10, H14
(c) (i)	3	9.5.4.3.2	H7, H12, H13, H14
(c) (ii)	2	9.5.4.2.1	H7, H14
(d) (i)	2	9.5.2.3.1	H2, H11
(d) (ii)	3	9.5.2.3.1	H2, H11, H14
(e)	7	9.5.3.2.2, 9.5.4.2.3, 9.5.5.3.5, 9.5.6.2.4	H4, H8, H13, H14

Section II

Question 34 — Shipwrecks, Corrosion and Conservation

Question	Marks	Content	Syllabus outcomes
(a)	3	9.6.4.2.2, 9.6.4.3.4	H3, H6, H14
(b) (i)	4	9.6.4.2.2, 9.6.4.2.4, 9.6.7.2.5	H7, H8, H13, H14
(b) (ii)	1	9.6.7.2.5	H4, H8, H14
(c) (i)	3	9.6.3.2.2	H7, H12, H13, H14
(c) (ii)	2	9.6.3.2.1	H7, H12, H13, H14
(d) (i)	2	9.6.4.3.2	H8, H11
(d) (ii)	3	9.6.4.3.2	H8, H11, H14
(e)	7	9.6.5.2.1, 9.6.5.2.2, 9.6.5.2.4, 9.6.5.3.2, 9.6.6.1, 9.6.6.2, 9.6.6.2.3	H1, H2, H8, H13, H14

Section II

Question 35 — The Biochemistry of movement

Question	Marks	Content	Syllabus outcomes
(a)	3	9.7.1.2.5, 9.7.8.2.1/5	H7, H9, H12, H14
(b) (i)	2	9.7.4.2.1, 9.7.4.2.2, 9.7.4.3.2.3, 9.7.4.3.1	H8, H9, H10, H13
(b) (ii)	3	9.7.4.2.6, 9.7.4.3.2, 9.7.4.2.5	H8, H14
(c) (i)	3	9.7.8.1, 9.7.8.2.5, 9.7.8.3.1, 9.7.10.2.1	H7, H8, H12, H14
(c) (ii)	2	9.7.9.2.3	H8, H7, H9, H13
(d) (i)	2	9.7.4.3.2	H8, H9, H11
(d) (ii)	3	9.7.4.3.2	H11, H14
(e)	7	9.7.2.1, 9.7.3.1, 9.7.6.1	H1, H4, H7, H8, H9, H14

Section II

Question 36 — The Chemistry of Art

Question	Marks	Content	Syllabus outcomes
(a)	3	9.8.4.2.3, 9.8.4.2.4	H6, H12, H13, H14
(b) (i)	2	9.8.3.2.1, 9.8.3.2.4	H6, H13, H14
(b) (ii)	3	9.8.3.2.4, 9.8.3.2.6, 9.8.4.2.3	H2, H6, H12
(c) (i)	2	9.8.2.2.4, 9.8.2.3.2	H2, H3, H6, H7, H11
(c) (ii)	3	9.8.2.2.5, 9.8.2.3.2	H2, H6, H7, H13
(d) (i)	2	9.8.2.2.1, 9.8.2.3.1	H6, H7, H11
(d) (ii)	3	9.8.2.2.1, 9.8.2.3.1	H11, H14
(e)	7	9.8.2.1, 9.8.2.2.4, 9.8.2.2.9	H3, H6, H7, H11, H14

Section II

Question 37 — Forensic Chemistry

Question	Marks	Content	Syllabus outcomes
(a)	3	9.9.2.2.2, 9.9.2.2.3	H6, H9, H13
(b) (i)	2	9.9.3.2.3, 9.9.3.2.4	H9, H10, H13
(b) (ii)	3	9.9.3.2.5, 9.9.3.3.2, 9.9.3.3.3, 9.9.3.3.4	H8, H9, H14
(c) (i)	4	9.9.4.2.3	H8, H9, H11, H14
(c) (ii)	1	9.9.4.2.2	H1, H12, H14
(d) (i)	4	9.9.1.2.1, 9.9.1.3.4, 9.9.1.3.3, 9.9.2.3.1, 9.9.3.3.2	H8, H9, H11, H14
(d) (ii)	1	9.9.5.2.1	H8, H9, H11
(e)	7	9.9.3.2.6, 9.9.3.3.5, 9.9.5.2.2, 9.9.5.2.3, 9.9.6.3.1	H3, H4, H11, H14